

NRES 746 – Workout of the Day #2: Pseudocoding a Permutation Test

Instructions: Work alone for about 5 minutes on Parts 1-2. Then get together with your neighbor(s) and spend about 5 minutes on Part 3, translating your pseudocode into actual R syntax **by hand, on paper** (no laptops).

Background: what is a permutation test? _____

In previous stats classes and in Lecture 1, you've seen tests for whether a sample's mean differed from a known null distribution: e.g., a z-test, and a t-test. These tests rely on knowing (or assuming) the shape of the population distribution the data came from.

A **permutation test** drops that requirement. It's a *nonparametric* (distribution-free) test: instead of simulating data from an assumed distribution, we generate a null distribution for our test statistic by **randomly reshuffling the labels** (e.g., treatment vs. control) on the data we already have, over and over. If the group labels really don't matter, shuffling them shouldn't change/reduce the expected difference between group means very much. If the observed difference is much larger than what typical reshufflings produce, that's evidence the labels *do* actually matter.

Part 1: The scenario _____

Recall the pygmy short-horned lizard example from Lecture 1: we want to know whether body mass differs between lizards in a cheatgrass-removal treatment (trtA) and a control group (Control).

trtA:	175	168	168	190	156	181	182	175	174	179
Control:	185	169	173	173	188	186	175	174	179	180

1a. What is the observed test statistic here (the single number we'll compare against the null distribution)?

1b. Is the alternative hypothesis one-tailed or two-tailed in this case? Briefly explain why.

Part 2: Write the pseudocode _____

2. Without looking back at your notes, write out – in plain English, step by step – the algorithm you would need to run a permutation test on this data. Think about: what gets shuffled, what gets recomputed each time, how many times you repeat it, and how you turn the result into a p-value.

Part 3: Translate to R (group, pen and paper) _____

3. As a group, fill in the blanks below to turn your pseudocode into real R syntax. You don't need working code – just your best attempt at the correct functions and syntax, written by hand.

```
permutation_test <- function(data, group, value, reps){  
  
  observed_dif <- _____  
  
  null_difs <- _____  
  
  for(i in 1:reps){  
  
    shuffled_labels <- _____  
  
    null_difs[i] <- _____  
  
  }  
  
  p_value <- _____  
  
  return(p_value)  
}
```

3a. Which R function did your group use to do the actual shuffling?

3b. What was the trickiest part of going from pseudocode to R syntax? Flag it – we'll come back to it!

